Behique status

Last updated 2026-10-28. Hackathon deadline: 2026-11-04, 9PM ET (Meta Global AI Developer Hackathon, Cerebral Valley). Track (Indie or Startup) and the official judging criteria are **not verified yet**.

An autonomous visual QA agent where a bug only counts if a replayable assertion proves it. The demo spine is a two-release regression check: a known good release, a release candidate, an evidence-backed report, and a human who makes the ship or hold decision.

1. Built and verified

Explorer (src/, earlier phases)

- Explores an unseen web app (screenshot, accessibility tree, network log), reports defects as assertions (data, not model-written code), and replays each one on a fresh app.
- Self-check rejects ungrounded reports. Also built: stability check, Playwright export, BM25 retrieval, token and cost accounting.
- Benchmark: todo-app with 28 planted bugs (27 scored). Run with `node bench.mjs --model baseten|meta`.
- **Caveats:** results are n=1 and noisy. Tuning test on Meta, 6 bugs: default 4/6, low 3/6, high 1/6, temperature 0 0/6, one action 2/6. Nothing beat the defaults. *reasoning low* is about 30% cheaper and 40% faster per step but hits the step cap more often.

Release pipeline (src/release/, built 2026-10-28, no model calls)

File	Job
build.mjs, adapters/freshrss.mjs	Check out a git ref (identified by sha), wipe data, install, seed one local feed, serve on a fixed port. About 2 s per build.
facts.mjs	Page facts (element path to attribute or visible text) after every step. Run-twice masking, diff, and ranking (changed, added, removed, moved).
compare.mjs	Record the base build twice, the candidate once, and the base a third time as a control.
diff-builds.mjs	Command line for the above: `node src/release/diff-builds.mjs 1.29.1 1.30.0`
confirm.mjs	Turns the top changed fact into a Playwright test and runs it 3 times per build on freshly seeded builds: `node src/release/confirm.mjs 1.29.1 1.30.0 [--bisect]`
scenarios.mjs	Named cases the checker can run (share, scroll): a feed fixture plus a few user settings, the clicks to replay, and an honest description stored with each report.
evidence.mjs	Evidence store: an append-only log where each event carries the fingerprint of the one before, plus evidence files named by their own fingerprint. Conclusions are labelled claims, and who may write each label is enforced: only code can write a verified fact, reproduced failure or measurement; only an AI model can write an AI guess; only a person records Ship or Hold. A claim must cite evidence that exists. Editing or removing anything earlier is detected.
hypothesis.mjs	The only paid step. Shows a model (Meta muse-spark-1.3) the observed difference and the one code change the search found, and nothing else, then records its answer as an AI guess. Code checks that the line the model points to really is in the change. Hard limits: 4,000 output tokens per call, 3 guesses per report, no automatic retry. The projected cost is shown before anything is spent.
bisect.mjs	Binary search over the commits between two versions, using the test. Each commit is good (test passed), bad (failed on the value itself) or skip (could not be tested, set aside and never guessed). Verified: the commit before passes 3/3, the first bad one fails 3/3.

Also added: an `attr` assertion kind in assert.mjs and export.mjs (not offered to the explorer prompt), and an optional start path in the exporter.

Report page (`/report`): the verdict with the logo, what the check proved (each claim links to its evidence), the code change found, a separate section for leads that are not proof (suspected causes and AI guesses, with an Ask an AI card that shows the projected cost and the exact text first), Ship and Hold buttons with a name and note (written to the record; a later decision does not erase an earlier one), a record-intact banner, the evidence files and the full event log.

Demo page: demo/server.mjs and demo/index.html (server name `demo`, port 8790). One story: an idea strip, a single Run the check button with a live timeline (including the bisect, about 90 s in all) and a verdict, a table of where the AI fits, a coming-soon list, and an optional hand-test section.

Result: FreshRSS 1.29.1 (good) vs 1.30.0 (bad), share-menu flow

Measure	Result
Facts recorded	1,090 stable, 174 volatile (13.8% masked)
Control run (good build, third time)	0 differences, so the masking can be trusted
Candidate differences	2 changed, 18 added, 0 removed, 21 moved (before collapsing moves: 2 / 39 / 21)
The 2 changed facts	The share button data-url and data-title are now URL-encoded. This is the real regression.
Exported test	**Passes 3/3 on 1.29.1, fails 3/3 on 1.30.0.** It fails on the value mismatch itself, not on a missing element.
First bad commit (bisect driver)	1b7178a1, "Security: apply encodeURIComponent to share template (#9146)", one file (p/scripts/main.js). 8 steps over 227 commits, verified 3/3 and 3/3, about 55 to 70 s. Same answer as the earlier throwaway script.

Second scenario: FreshRSS #9285, scroll position

This is not a confirmed bug. The maintainers could not reproduce it as reported, the reporter closed it himself, and no fix exists. The behaviour was changed on purpose in #8870. It is a real, reproducible difference, so it is shown as a behaviour change for a person to judge, not as a bug.

Measure	Result
Setup	8 long articles, Stick the article to the top when opened switched off. Open article 1, bring article 2 into view, click article 2.
What differs	1.29.1 does not scroll (0 px). 1.30.0 scrolls to the top (4,033 px higher). Exactly one changed fact; control run 0 differences.
Exported test	Relative: the scroll change across the last click. Passes 3/3 on 1.29.1, fails 3/3 on 1.30.0.
First bad commit	aaae9ee3, "Always jump article to top when header is offscreen, also when Stick the article to the top when opened is disabled (#8870)". 8 steps, verified 3/3 and 3/3, about 100 s.

2. Not built, in order

#	Piece	What it is
1	Explorer finding the Share flow by itself	NOT achieved. Without a hint it missed it 2 of 2 times (about $0.19 and $0.22): it never clicked an article, and Share only exists inside an expanded article. WITH A HINT in the brief ("some controls only appear after you open an item", no mention of Share) it found it by step 3 (1 of 1 hinted run, about $0.28). A hinted result is not the same as finding it unaided. Its own path, replayed through the release checker, reproduces the regression. Not done: wiring explorer steps into the demo, or trying a prompt change instead of a hint.
2	Demo recording	Last.

3. Known limits

- The test finds the element by its position in the page (XPath), so it only works on builds with the same markup path.
- A changed value inside a moved element still shows as removed plus added with different values.
- Steps are hand-written (bench/fixtures/freshrss/steps.json), not generated by the explorer.
- The real clipboard copy is verified (headless Chromium): 1.29.1 copies the normal link, 1.30.0 the double-encoded one.
- The repo is not under git, so edits are not reversible. Run `git init` and commit before the next piece.

4. Open questions

- Verify the official hackathon rules, judging criteria, and whether Indie or Startup fits.
- Optional runs not done: repeat no-retrieval noise run (about $5.3) and *reasoning low* with 30 steps on all 27 bugs (about $4). Skip until the pipeline is done.
- The baseline artifact (V6skE3HaNcVDwV5sV57ZdU) is out of date.

5. Credits (estimates)

Source	Amount	Note
Meta Model API	$150, about $146 left	Per the last check.
Baseten	$100, about $70 left	The key only works with meta-models/muse-spark-open-1026.
Modal, Fireworks	unconfirmed	
Spent so far	about $38	Of about $250 in credits, by estimate. The first AI guess added about $0.006 and the three explorer runs on FreshRSS about $0.69.

6. Where things are

Repo	~/dev/behique (CLAUDE.md has layout and rules; docs/STATUS.md and docs/regression-demo-plan.md)
FreshRSS worktrees and feed	~/dev/freshrss-spike (PHP via Homebrew)
Leviathan (read-only reference)	~/dev/leviathan
Exported test	out/regression/1.29.1-vs-1.30.0/R1.spec.js